

Career Compass: AI Virtual Career Counsellor

Introduction

In today's competitive job market, students and professionals often face confusion while choosing the right career path based on their skills. This project aims to build an AI-powered virtual career counsellor that helps users identify top trending career opportunities in various engineering domains, based on their skill sets. The application mimics the decision-making of a career counsellor using machine learning and skill-to-role matching.

Abstract

The AI Virtual Career Counsellor is a web-based application built using Python (Flask) and machine learning techniques. It takes a user's input skills and predicts the most suitable career options within a selected domain (e.g., AI, Mechanical, ECE, etc.). It utilizes datasets related to job roles and skill trends, analyzes trending scores, and recommends job titles with the highest match. The model identifies skill gaps and provides a career trajectory by highlighting required additional skills.

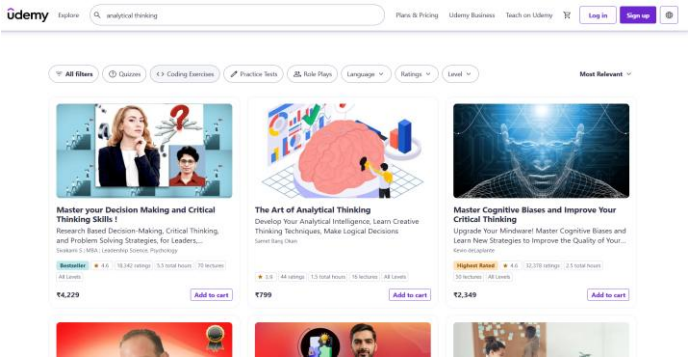
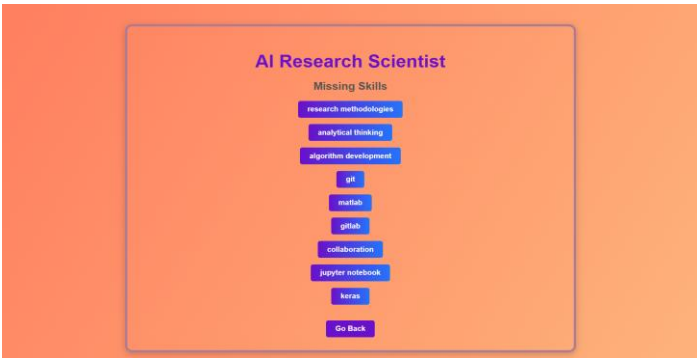
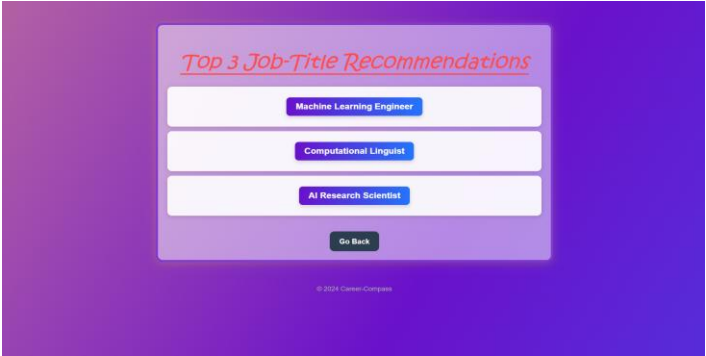
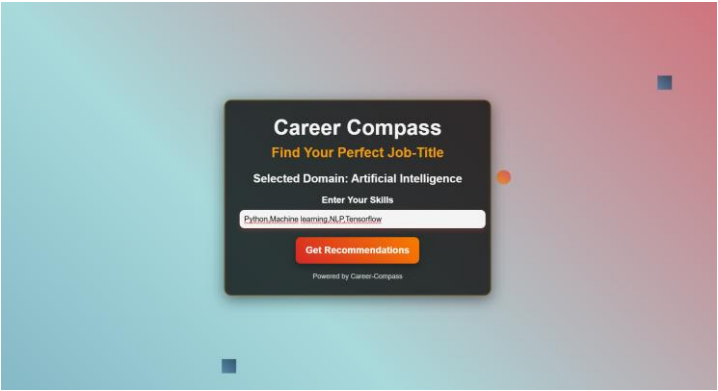
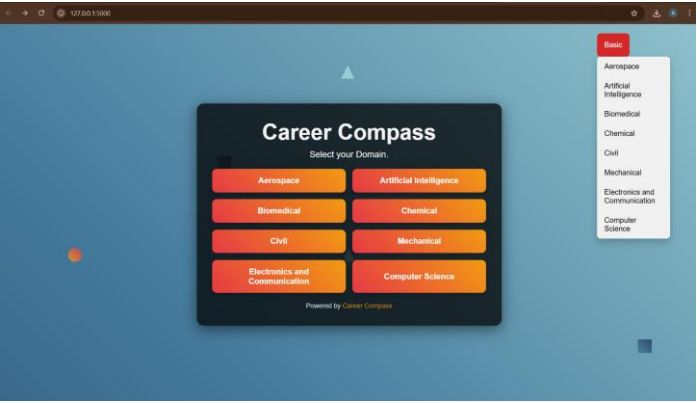
Tools Used

- Python 3.x
- Flask
- Pandas
- Scikit-learn
- HTML/CSS (Frontend)
- WTForms (User Input)
- Jinja2 (Template rendering)

Steps Involved in Building the Project

1. Dataset Preparation: CSVs per domain (jobs and trends) merged by JobTitle.
2. Preprocessing: Skill columns combined, trending scores filled.
3. Model: Logistic Regression trained on combined skills vs trending scores.
4. Input: Users input skills via form.
5. Matching & Prediction: Skills are matched, score calculated using skill match, trend score, and ML prediction.
6. Web UI: Users select domain input skills get top 3 job suggestions with missing skills if any.

Results:



Conclusion

The AI Virtual Career Counsellor helps users discover suitable career options by analyzing current skills and job trends. It provides clear guidance for perfect-fit roles and outlines the missing skills required for trending job titles. This tool is especially useful for students, institutions, and job-seekers looking for AI-driven career advice.